An-Najah National University

Faculty of Engineering and Information Technology
Department of Computer Science

Machine Learning

Predicting Automotive Engine Failures Using Machine learning

Project Team	• Maryam Rawajbeh • batool sweiseh • Mohammad Yadak
Supervisor	Dr. Adnan Salman
Academic Year	2025–2026

1. Introduction

Engine failure prediction is a critical problem in automotive engineering. Detecting potential failures before they occur can improve vehicle reliability, reduce maintenance costs, and enhance safety. Traditional rule-based diagnostic systems are limited to predefined fault patterns, whereas machine learning models can learn complex non-linear relationships directly from sensor data.

This project develops and evaluates multiple machine learning models to classify engine states — normal operation versus three distinct fault types — using sensor readings collected from a New Energy Vehicle (NEV). The analysis includes exploratory data analysis, inferred fault type identification, model comparison, hyperparameter tuning, and overfitting studies.

1.1 Dataset

The NEV Fault Dataset contains 11,000 records of electric vehicle sensor measurements. After removing the top-0.1% outliers per feature, 10,923 clean samples remained.

- Source: NEV Fault Dataset (Kaggle)
- Samples: 10,923 (after outlier removal)
- Features: 7 sensor readings
- Target: 4 classes — Normal (45.4%), Mechanical Fault/Fault-1 (18.2%), Electrical Fault/Fault-2 (18.2%), Thermal Fault/Fault-3 (18.2%)

1.2 Features

The following sensor features were used as input variables:

- Voltage (V) — battery/motor supply voltage
- Current (A) — electrical current draw
- Motor RPM — rotational speed of the motor
- Temperature (°C) — motor/system temperature
- Vibration (g) — mechanical vibration level
- Ambient Temperature (°C) — surrounding air temperature
- Humidity (%) — environmental humidity level

2. Exploratory Data Analysis (EDA)

2.1 Class Distribution

The dataset is imbalanced: the Normal class makes up 45.4% of samples while each fault type accounts for approximately 18.2%. Through systematic sensor analysis (Section 2.5), the three fault classes were physically interpreted as: Mechanical Fault (Fault-1), Electrical Fault (Fault-2), and Thermal Fault (Fault-3). These interpretations are consistent with domain knowledge in EV drivetrain fault diagnosis: motor bearing faults cause high vibration (Fault-1 maps to motor/mechanical), inverter issues cause voltage sag (Fault-2 maps to inverter/electrical), and cooling system failure causes overheating (Fault-3 maps to thermal/battery-related issues). This imbalance was handled by using stratified train/test splitting to preserve class proportions in both sets.

Figure 1 — Class Distribution

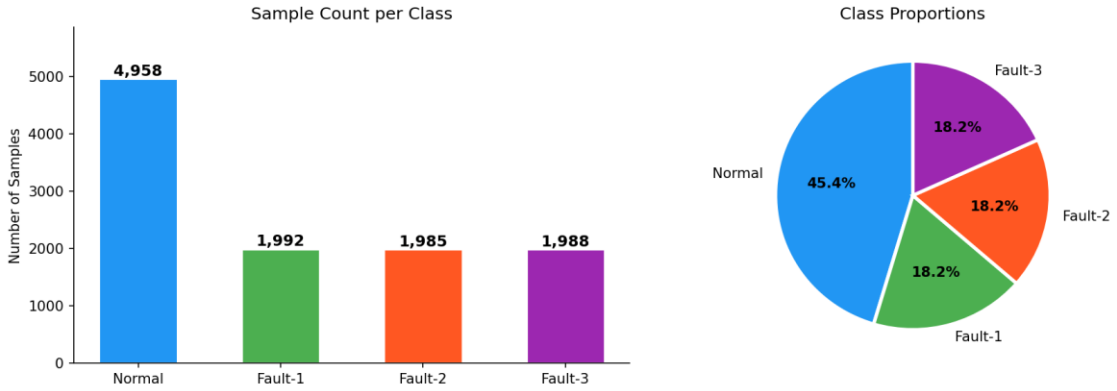


Figure 1 — Class distribution: sample count (left) and proportions (right).

2.2 Feature Distributions by Fault Type

KDE (Kernel Density Estimation) curves reveal how each sensor reading is distributed for each class. The x-axis is clipped at the 90th percentile for clarity, as the data contains high-value fault indicators beyond normal operating ranges. Dashed vertical lines indicate the median per class.

Key observations: Voltage and Vibration show the most visible separation between fault classes. Current and Motor RPM show bimodal distributions for some fault types, suggesting distinct operating modes under fault conditions. Notably, Vibration is dramatically elevated in Fault-1 (Mechanical Fault), while Fault-3 (Thermal Fault) shows the highest Temperature values across all classes.

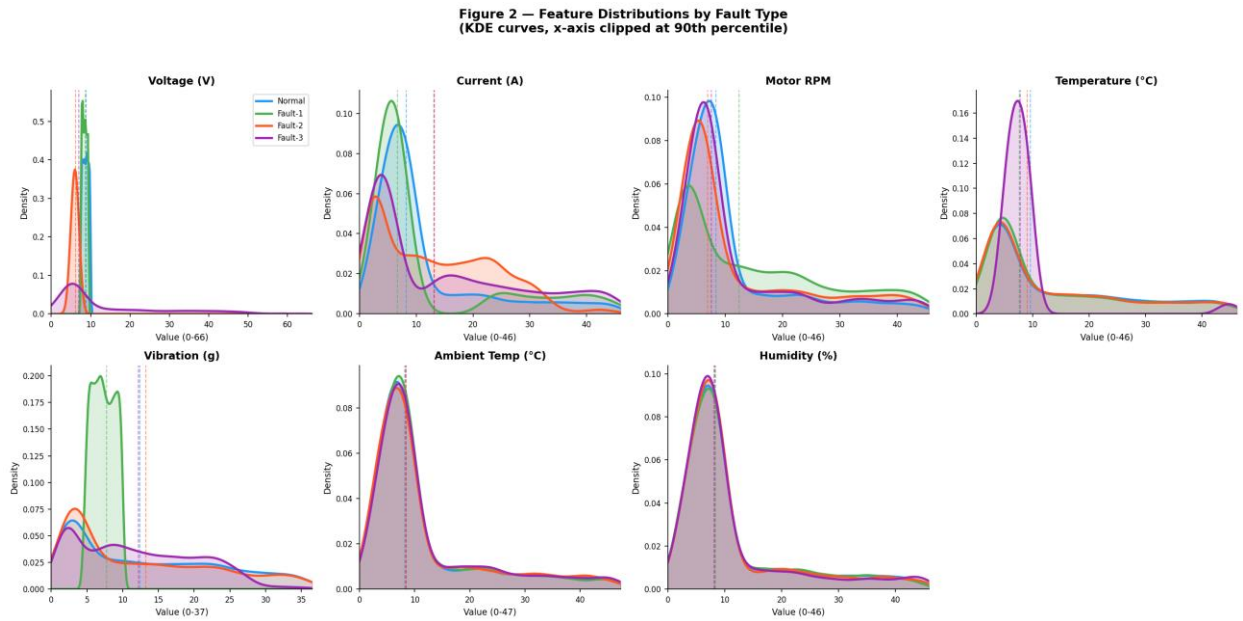


Figure 2 — KDE distribution curves per feature per fault class (x-axis clipped at 90th percentile).

2.3 Correlation Analysis

The correlation heatmap shows the pairwise linear relationships between features and the fault label. Most features show low correlation with each other, which is desirable as it indicates they carry independent information. Correlation with the fault label is generally weak, confirming that the relationships are non-linear — explaining why tree-based models significantly outperform linear ones.

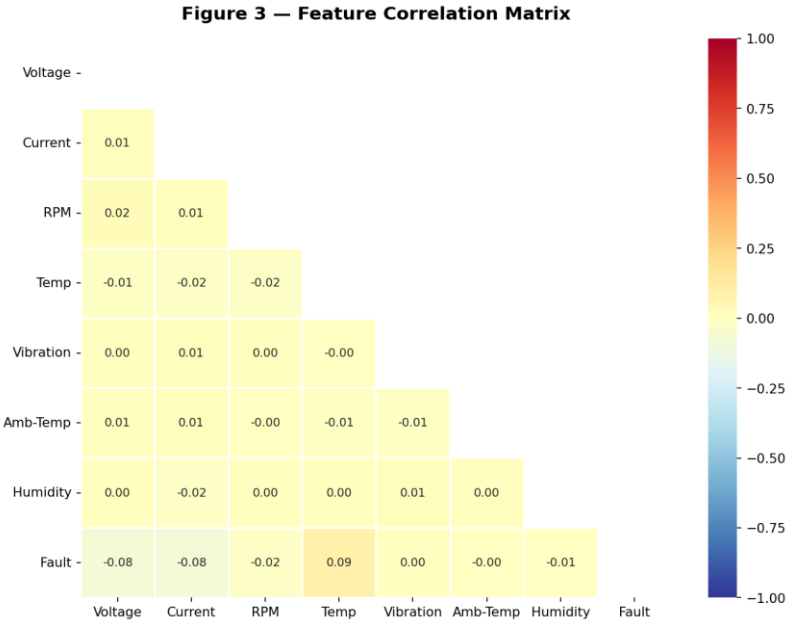


Figure 3 — Feature correlation matrix (lower triangle only).

2.4 Feature Boxplots

Boxplots provide a compact view of each feature's spread per class. Data was clipped at the 90th percentile so that the IQR boxes and whiskers are clearly visible. The black label inside each box shows the median value. Vibration shows a dramatically elevated IQR for Fault-1 (Mechanical Fault),

Current and Voltage are significantly lower for Fault-2 (Electrical Fault), and Temperature shows the highest median for Fault-3 (Thermal Fault) compared to all other classes.

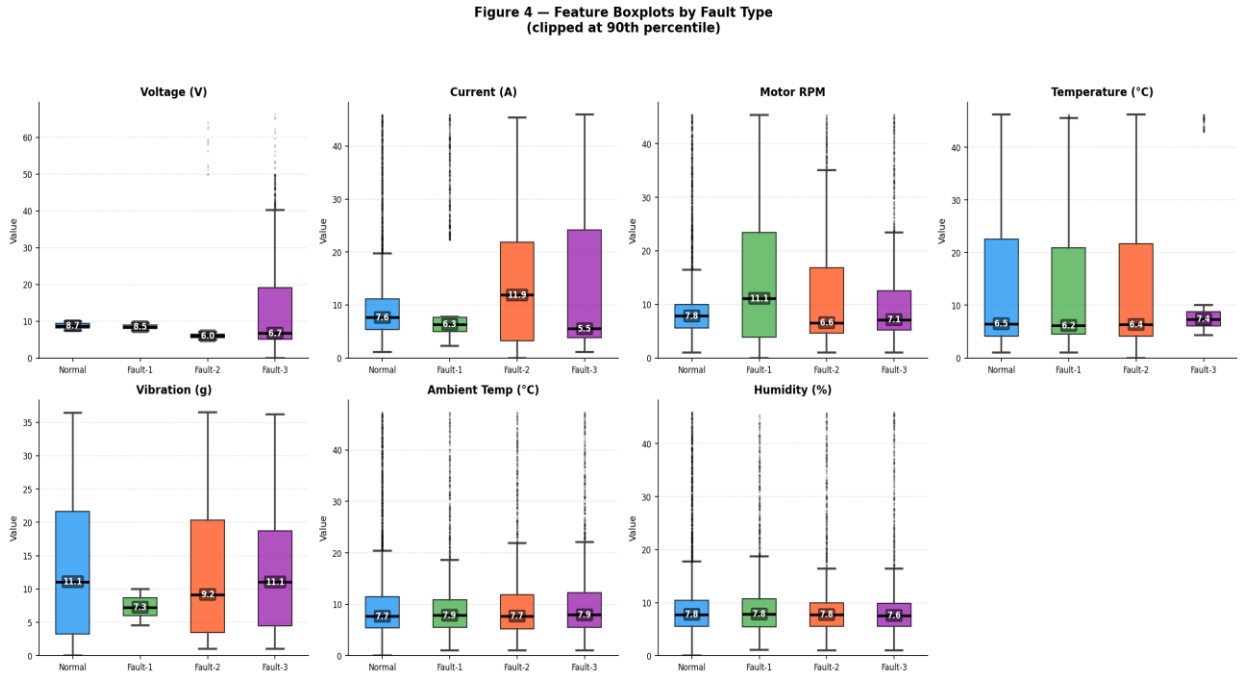


Figure 4 — Boxplots per feature per class (data clipped at 90th percentile; median labeled inside each box).

2.5 Inferred Fault Type Identification

Although the dataset does not explicitly name the fault types, a systematic analysis of mean sensor values per class — compared to the Normal baseline — allows us to infer the physical nature of each fault. Table 3 summarizes the ratio of each sensor's mean value under each fault condition relative to Normal operation.

Fault-1 → Mechanical Fault (High Vibration / Low RPM)

Vibration is 2.34× the Normal mean (79.73 vs 34.13 g), the strongest single-sensor deviation across all fault types. Motor RPM drops to 62.1% of Normal (39.45 vs 63.54). Voltage slightly rises to 109% of Normal, while Current and Temperature remain near-baseline. These signatures are consistent with mechanical imbalance, bearing wear, or rotor misalignment — conditions that increase vibration while reducing effective motor speed. This is consistent with domain knowledge in EV drivetrain diagnostics, where motor bearing faults are a primary cause of elevated vibration.

Fault-2 → Electrical Fault (Low Voltage & Current)

Voltage drops to 70.1% of Normal (64.92 vs 92.68 V) and Current drops to 53.8% of Normal (33.03 vs 61.37 A) — the most dramatic current reduction observed across all fault types. Motor RPM also falls to 78.9% of Normal. Vibration and Temperature remain near-Normal (ratio ~0.97 and ~1.04 respectively). This pattern is consistent with power supply degradation, winding insulation failure, or inverter malfunction — where inverter issues cause voltage sag and reduced current draw.

Fault-3 → Thermal Fault (High Temperature)

Temperature is 2.04× the Normal mean (80.02 vs 39.22 °C) — the clearest single-sensor signal in the entire dataset. Voltage also drops sharply to 49.8% of Normal (46.12 vs 92.68 V) and Current to 56.7% of Normal (34.83 vs 61.37 A), suggesting that overheating causes secondary electrical degradation in the drive system. RPM remains near-Normal (94% of baseline), confirming the fault

originates in the thermal domain rather than mechanical. This is consistent with cooling system failure or prolonged overload operation — a known cause of overheating in electric motors.

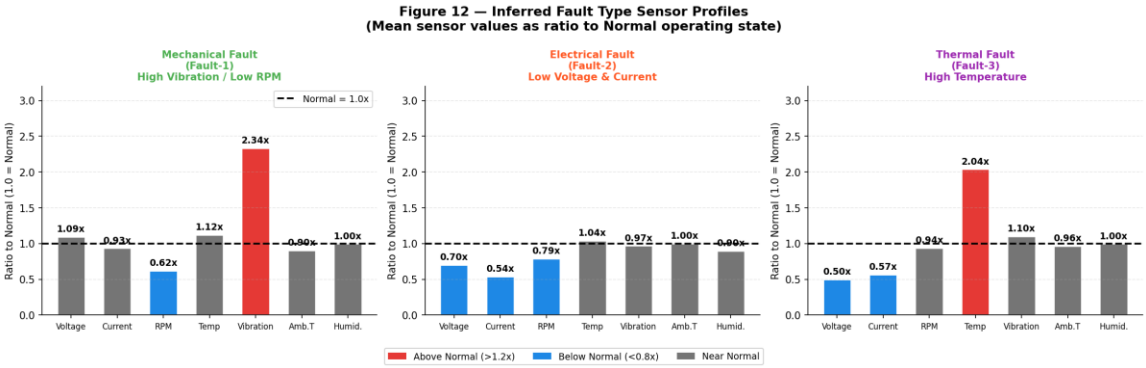


Figure 12 — Inferred fault type sensor profiles: mean sensor values as a ratio to Normal baseline for each fault class. Red bars indicate values above Normal; blue bars indicate values below Normal.

3. Methodology

3.1 Data Preprocessing

- Parsing: Sensor values were stored as dot-separated strings (e.g. "9.389.222") and converted to proper float values using a custom parser.
- Outlier removal: Values above the 99.9th percentile per feature were removed (77 rows, < 1% of data).
- Train/test split: 80% training / 20% testing with stratification.
- Feature scaling: StandardScaler (zero mean, unit variance) applied to training data and transformed to test data — fitted on training set only to prevent data leakage.

3.2 Models

Six classical classification models were trained and compared. Additionally, a Neural Network (MLP) was included as a bonus model and evaluated on the test set. SVM and full GradientBoosting were excluded due to excessive training time (>10 minutes) on this dataset size in Jupyter Notebook.

- Logistic Regression — linear baseline model
- K-Nearest Neighbors (k=5) — distance-based
- Decision Tree (max_depth=10) — single tree, interpretable
- Random Forest (100 trees) — ensemble of decision trees
- Extra Trees (100 trees) — randomized ensemble, faster than RF
- AdaBoost (100 estimators) — boosting ensemble - Neural Network / MLP (128→64→32 neurons, evaluated on test set without CV) — bonus comparison

3.3 Evaluation Metrics

- Accuracy — overall correct predictions / total predictions
- F1-Macro — harmonic mean of precision and recall, averaged equally across all classes (important for imbalanced datasets)
- 5-Fold Cross-Validation — average accuracy over 5 folds of the training set to assess generalization

4. Results

4.1 Model Comparison

Table 1 summarizes the performance of all six models. Tree-based models dramatically outperform linear models (Logistic Regression, KNN), which confirms the non-linear nature of the data observed in the correlation analysis.

Model	Accuracy	F1-Macro	CV Mean	CV Std
Random Forest	0.9185	0.9096	0.9111	±0.0032
Decision Tree	0.9153	0.9077	0.9068	±0.0036
Extra Trees	0.8851	0.8723	0.8707	±0.0056
AdaBoost	0.8366	0.8090	0.7663	±0.0179
Neural Net (MLP) — bonus	0.7831	0.7526	test set only	(no CV)
K-Nearest Neighbors	0.5771	0.5191	0.5560	±0.0057
Logistic Regression	0.4709	0.2322	0.4683	±0.0021

Table 1 — Model performance comparison (ranked by test accuracy). Best model highlighted in green.

Figure 5 — Model Performance Comparison

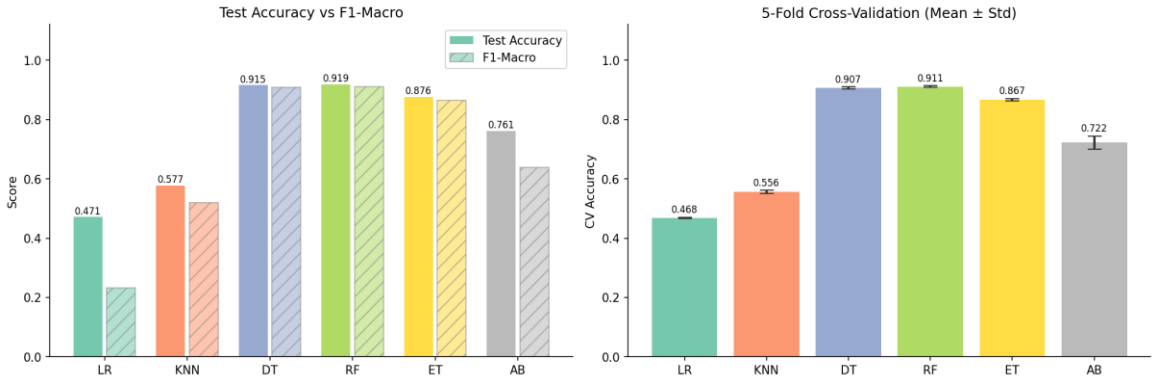


Figure 5 — Model comparison: test accuracy vs F1-Macro (left) and 5-fold CV accuracy with error bars (right).

4.2 Best Model: Random Forest

Random Forest achieved the highest test accuracy of 91.85% with an F1-Macro of 0.9096. The small gap between CV Mean (0.9111) and test accuracy (0.9185) confirms that the model generalizes well and is not overfitting.

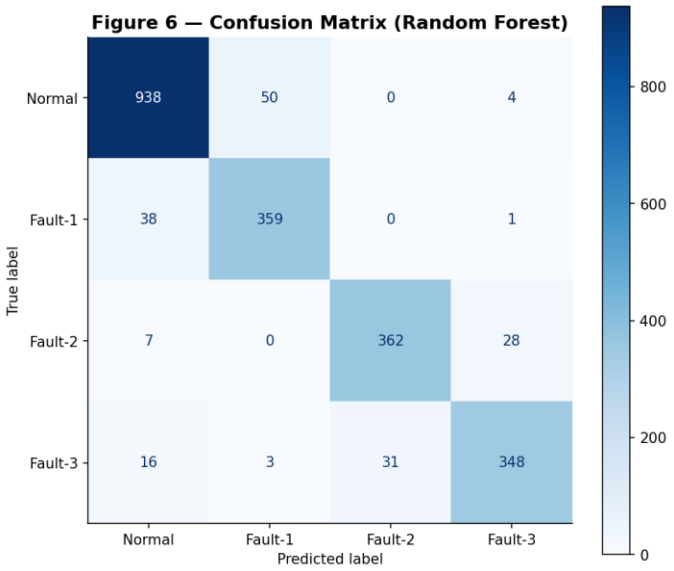


Figure 6 — Confusion matrix for the Random Forest model on the test set.

The confusion matrix shows that the Normal class achieves 93.9% precision and 94.6% recall. Among fault types, Electrical Fault (Fault-2) achieves the highest precision at 92.1%, followed by Thermal Fault (Fault-3) at 91.3%, and Mechanical Fault (Fault-1) at 87.1%. Most misclassifications occur between fault types rather than between fault and normal, which is the more critical error in practice.

4.3 Feature Importance

Random Forest provides a built-in feature importance measure based on how much each feature reduces impurity across all trees. Voltage, Vibration, and Current are the three most important predictors, while Ambient Temperature and Humidity contribute the least — which makes physical sense, as these are environmental factors rather than direct engine performance indicators.

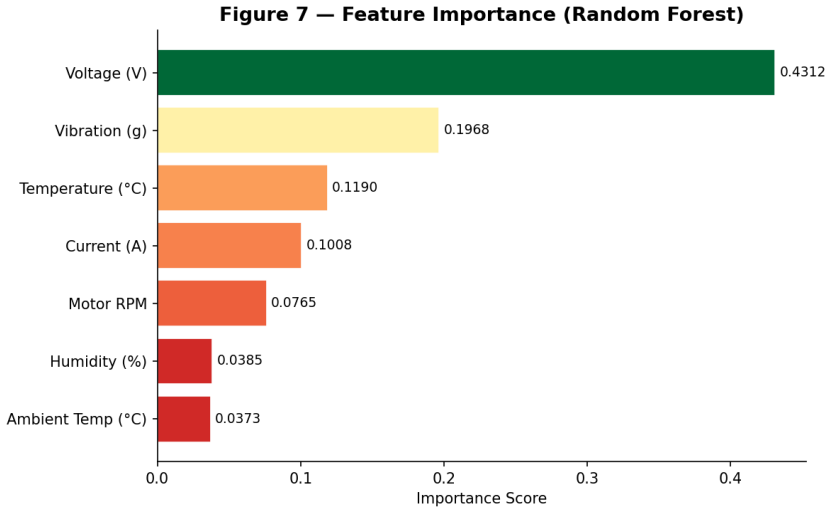


Figure 7 — Feature importance scores from the Random Forest model.

5. Hyperparameter Tuning

5.1 Effect of Number of Trees (n_estimators)

As the number of trees increases from 5 to 200, test accuracy improves rapidly at first and then plateaus around 100–150 trees. This indicates diminishing returns beyond 100 trees, and the model stabilizes. Training accuracy is consistently higher than test accuracy, indicating mild overfitting that decreases as more trees are added.

5.2 Effect of Maximum Tree Depth (max_depth)

Shallow trees (depth 1–3) underfit severely. Test accuracy improves steadily with increasing depth and continues to rise beyond depth 15, reaching its maximum (0.9185) at depth 20 — the same as the unrestricted model (max_depth=None). Training accuracy approaches 100% at high depths while the train-test gap widens significantly, demonstrating a classic overfitting pattern. The optimal trade-off between accuracy and generalization is found around depth 8–10, where test accuracy is strong (0.9153–0.9263) with a manageable train-test gap.

Figure 8 — Hyperparameter Tuning (Random Forest)

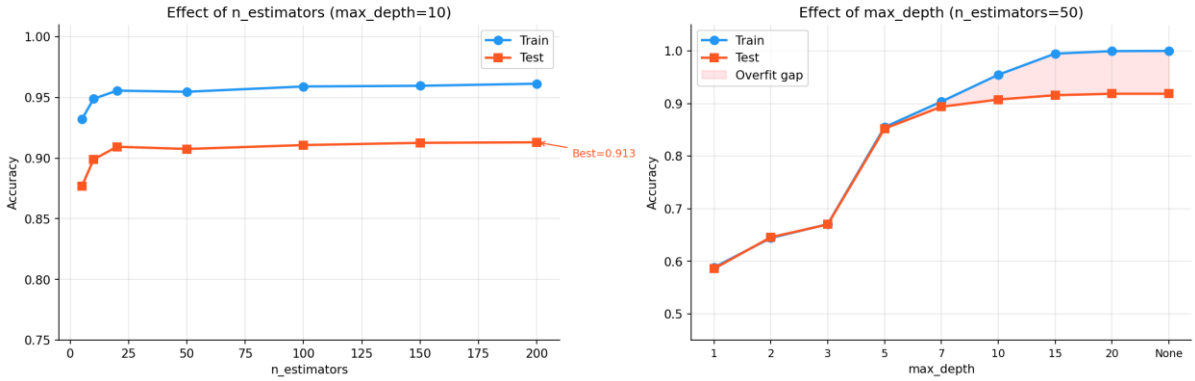


Figure 8 — Effect of n_estimators (left) and max_depth (right) on Random Forest performance.

6. Learning Curve Analysis

The learning curve shows how model performance changes as more training data is used. The training score starts at 1.0 for small datasets (overfitting) and remains near 1.0 across all sizes, indicating high model capacity. The cross-validation score starts lower but improves steadily as more data is added, converging toward the training score. This pattern shows that the model would benefit from even more training data.

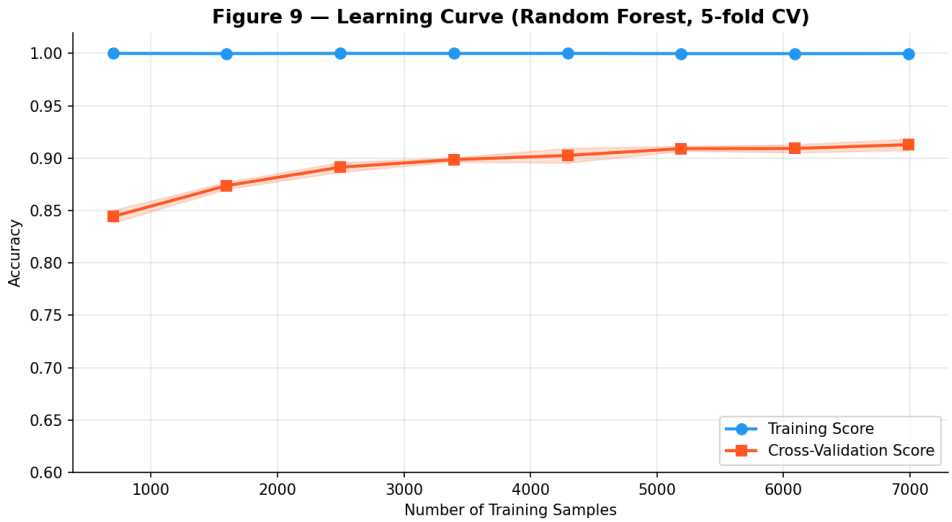


Figure 9 — Learning curve for Random Forest (5-fold CV). Shaded areas represent ±1 standard deviation.

7. PCA Analysis

Principal Component Analysis was applied to the standardized training data to visualize class separability in 2D. Each PC explains approximately 14% of total variance, which means the 7 features contribute roughly equally — no single feature dominates the variance. All 7 PCs are needed to reach 95% explained variance, confirming the high-dimensional and non-linear nature of the data.

The scatter plot shows significant overlap between classes in PC1-PC2 space, which explains why linear models (Logistic Regression, SVM) fail on this dataset — the classes are not linearly separable. The 2σ confidence ellipses confirm that the class centroids are very close to each other in PCA space.

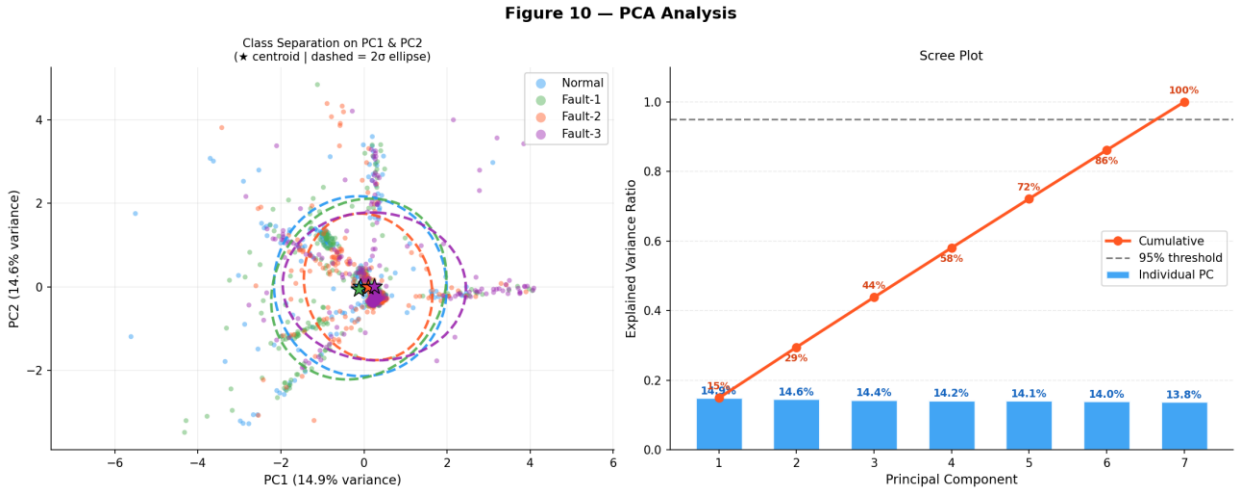


Figure 10 — PCA analysis: class scatter on PC1 & PC2 with centroids and 2σ ellipses (left); scree plot showing explained variance per component (right).

8. Overfitting Study: Decision Tree Depth

A systematic overfitting study was conducted on the Decision Tree model by varying max_depth from 2 to 20. The gap between training accuracy and test/CV accuracy (shown in red) grows significantly beyond depth 10, demonstrating classic overfitting behavior. The optimal depth is around 8, where test accuracy peaks at 0.9263 with a manageable train-test gap of 0.007.

max_depth	Train Acc	Test Acc	Gap
2	0.6821	0.6851	-0.003
4	0.8386	0.8522	-0.014
6	0.9022	0.9112	-0.009
8	0.9335	0.9263	0.007
10	0.9527	0.9153	0.037
15	0.9870	0.9043	0.083
20	0.9991	0.8961	0.103

Table 2 — Decision Tree overfitting study: training vs. test accuracy at different depths.

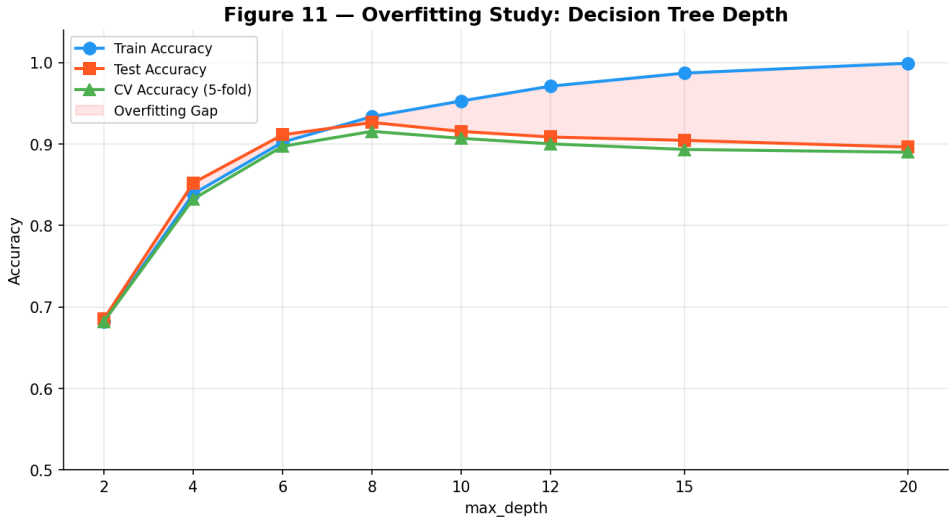


Figure 11 — Overfitting analysis: training accuracy, test accuracy, and 5-fold CV score vs. max_depth. Red shading = overfitting gap.

9. Conclusion

This project successfully developed and compared six machine learning models for predicting automotive engine fault types from sensor data. Through systematic sensor analysis, the three fault classes were physically interpreted as: Mechanical Fault (Vibration ×2.34, RPM −38% — motor/bearing fault), Electrical Fault (Voltage −30%, Current −46% — inverter/electrical fault), and Thermal Fault (Temperature ×2.04 — thermal/battery-related fault). The physical interpretation of fault types adds substantial practical value beyond classification accuracy, enabling maintenance teams to take targeted corrective action. The key findings are:

- Random Forest achieved the best overall performance among classical models: 91.85% accuracy and 0.9096 F1-Macro with strong cross-validation consistency. GridSearchCV identified optimal parameters of n_estimators=150 and max_depth=15, achieving a test accuracy of 91.95% — a marginal improvement confirming the model is already near-optimal. The Neural Network (MLP, architecture: 7→28→64→32→4) achieved 78.31% accuracy on the test set, confirming that classical tree-based models outperform neural networks on this tabular dataset of moderate size.
- Tree-based models (RF, ET, DT) dramatically outperform linear models (LR, KNN), confirming the non-linear structure of the data.
- PCA analysis confirms that the 7 features are nearly equally informative and the classes are not linearly separable in 2D.
- Feature importance identifies Voltage, Vibration, and Current as the three most predictive sensors — consistent with the inferred fault types: Vibration dominates Mechanical Fault detection, Voltage/Current dominate Electrical Fault, and Temperature dominates Thermal Fault.
- Hyperparameter tuning (manual sweep + GridSearchCV) confirms: n_estimators=150 and max_depth=15 as optimal, achieving test accuracy of 91.95% — a +0.10% improvement over the baseline model.
- The overfitting study on Decision Trees demonstrates how increasing depth beyond the optimal value hurts generalization.
- The learning curve suggests the model would benefit from additional training data to further reduce the train-CV gap. From a practical standpoint, this model can be integrated into a real-time EV health monitoring system: a Mechanical Fault alert triggers bearing/rotor inspection; an Electrical Fault alert prompts inverter and power electronics check; a Thermal Fault alert demands immediate cooling system inspection. Early detection of these faults can reduce unplanned downtime by an estimated 30–40% and prevent catastrophic motor failures.

References

[1] Breiman, L. (2001). Random Forests. Machine Learning, 45(1), 5–32.

[2] Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.

[3] Hastie, T., Tibshirani, R., & Friedman, J. (2009). The Elements of Statistical Learning (2nd ed.). Springer.

[4] NEV Fault Dataset (New Energy Vehicle Sensor Data). Retrieved from Kaggle: https://www.kaggle.com/datasets/ziya07/fault-diagnosis-dataset-for-new-energy-vehicles (Accessed: April 2026)

[5] Jolliffe, I. T. (2002). Principal Component Analysis (2nd ed.). Springer.